



CENTRAL UNIVERSITY OF SOUTH BIHAR

Department of Computer Science

Class Test-1

Semester: 1st

Course title: Data Structure and Algorithms

Course code: CSC81DC00204

Maximum Points: 10

Duration: 1 hour

1. Let the base address of one dimensional linear array (LA) is 1000 and each element of the array occupies 4 bytes. Find the address of LA (41). **(2 marks)**
2. Convert following infix expressions to prefix expression:
 $(a+(b/c)*(d^e)-f)$ **(3 marks)**
3. Evaluate following postfix expression using stack:
 $10\ 5\ +\ 60\ 6\ /\ * 8\ -$ **(3 marks)**
4. Write two applications of queue data structure. **(2 marks)**



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Department of Computer Science

Lab Test

Course title: Data Structure and Algorithms
Maximum Points: 10

Semester: 1st
Duration: 1 hour

1) Write a C program to implement a stack using arrays of integers. Your program should perform **push** and **pop** operations on the stack:

OR

Write a C program that sorts a given list of integers in ascending order using bubble sort. The program should take the list of integers from the user through the keyboard.



CENTRAL UNIVERSITY OF SOUTH BIHAR
Department of Computer Science

End Term Examination

Academic Session: 2025-26

Semester: 1st

Programme: M.Sc. Computer Science

Duration: 3 hours

Course Code: CSC81DC00204

Course Title: Data Structure and Algorithms

Enrolment No. (to be filed by student):

Total Points: 70

There are three sections (A, B, and C). Answer all questions as per instructions. Points against each question are mentioned in the parenthesis.

Section – A

Answer all the questions in not more than 50 words. Each question carries 1 Point.

(1 x 10 = 10 Points)

1. Define a complete binary tree.
2. Write the core operations of an abstract data type (ADT).
3. Let the base address of a one-dimensional linear array (LA) be 4000 and each element of the array occupies 2 bytes. Find the address of LA (32). Assume that the lower bound is 1.
4. Write the names of various operations that can be performed on the Queue data structure.
5. Define height of a tree.
6. Write two differences between homogeneous and non-homogeneous data structures.
7. Define a minimum spanning tree.
8. Define a binary search tree with an example.
9. Write differences between graph and tree data structures.
10. How does linear data structure differ from non-linear data structure?

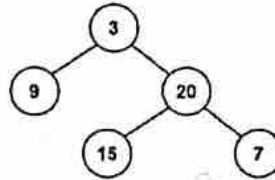
Section – B

Answer ANY FIVE questions in not more than 300 words. Each question carries 6 Points.

(6 x 5 = 30 Points)

1. Write an algorithm to insert an element into an array.
2. Evaluate following postfix expression using stack:
 $a b + c d / * e -$, where $a = 10$, $b = 5$, $c = 60$, $d = 6$ and $e = 8$.
3. Sort the following numbers: 40, 20, 30, 24, 12 using bubble sort.

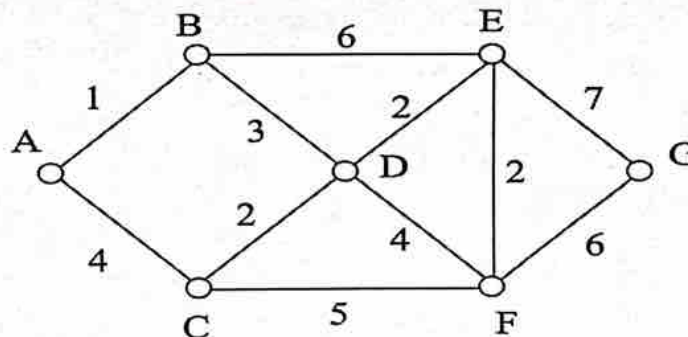
4. Convert infix expression $(a - b / c) / (e * d)$ to postfix expression.
5. Write an algorithm to insert an element into a queue.
6. Find preorder and postorder traversal of the following binary tree:



Section – C

Answer ANY TWO questions in not more than 1000 words. Each question carries 15 Points.
(15 x 2 = 30 Points)

1. Write a C program to implement the binary search algorithm. The input of the algorithm is a sorted list of integers and the element to be searched. The algorithm should give the location of the element as output if the element is found in the list otherwise it should print the message, "The element is not found in the list".
2. Find minimum spanning tree of the following graph using Kruskal and Prim's algorithms.



3. Consider the following array of integer values:

[20, 10, 5, 40, 12, 15, 25, 2, 45]

Sort the array using quick sort. Write clearly all intermediate steps.